Optimizing Transportation Cost for Travelling

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Introduction

1.1 What is Optimization?

Optimization is the process of finding the best solution to a problem from a set of possible choices.

It involves:

- Maximizing or minimizing a specific objective (like cost, profit, time, distance, etc.)
- Under given constraints (such as resources, time limits, or capacity).

1.2 Importance of Optimization in Real Life

Optimization plays a crucial role in improving efficiency, saving resources, and making better decisions in everyday situations. Here's how it impacts real life:

- Minimizes production cost by efficiently using raw materials and labor.
- Maximizes output and profit without increasing resource consumption.
- Optimizes water and energy usage, reducing waste

1.3 What is Linear Programming Problem (LPP)?

A Linear Programming Problem (LPP) is a mathematical method used to find the best possible outcome (such as maximum profit or minimum cost) in a situation with limited resources.

Key Elements of LPP:

- Objective Function:
- Decision Variables:
- Constraints:
- Non-negativity Conditions:

1.4 Historical Background and Applications of LPP

Linear Programming (LP) originated from the need to optimize resource allocation during times of constraint—especially during World War II.

1940s (WWII Era):

- The concept of LP was developed by George B. Dantzig (an American mathematician). He created the Simplex Method in 1947, a powerful algorithm to solve linear programming problems.
- It was used by the military to plan logistics like minimum-cost supply routes and fuel usage

Linear Programming is used in many fields to maximize output or minimize cost with limited resources. Here's where and how it's applied:

- Manufacturing: Optimize production schedules and resource allocation.

  Example: Maximize the number of products made with limited raw materials and labor.

- Transportation and Logistics Minimize transportation cost and delivery time.

  Example: Best way to deliver goods to multiple cities with the least fuel used.

- Finance and Investment Maximize return on investment with given budget and risk constraints.

  Example: Choosing a mix of stocks and bonds that gives the best return.

Optimization Problem

2.1 Problem Statement

A travel company is organizing transportation for tourists using two travel routes, **Route A** (x) and **Route B** (y). They need to decide how many vehicles to assign to each route so that the total transportation cost is minimized, while ensuring all travel requirements are satisfied.

- The company wants to minimize the total travel cost.
- There is a minimum traveler capacity.
- There is a maximum vehicle availability.

2.2 Decision Variables

Let:

- x = number of vehicle units assigned to Route A
- y = number of vehicle units assigned to Route B

2.3 Objective Function

Minimize the total travel cost:

$$\text{Minimize } Z = 4x + 5y$$

2.4 Constraints

$$\begin{aligned} 2x + 3y &\geq 6 && \text{(Minimum traveler capacity)} \\ 3x + y &\leq 3 && \text{(Maximum vehicle availability)} \\ x, y &\geq 0 && \text{(Non-negativity)} \end{aligned}$$

2.5 Solution Methods

This problem can be solved using:

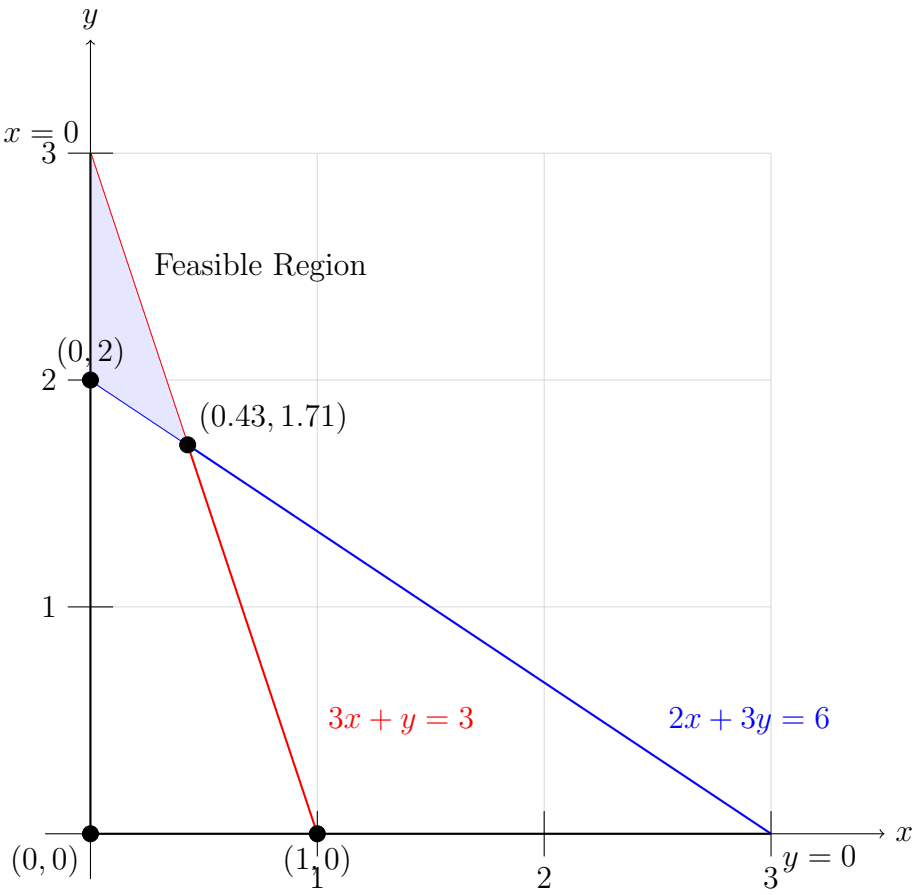
- **Graphical Method**, as there are only two decision variables.
- **Simplex Method (Two-Phase)**, for a more generalized algebraic approach.

Graphical Method

* Objective: Minimize $Z = 4x + 5y$
* Constraints:

$$\begin{aligned} 2x + 3y &\geq 6 \quad \text{(Demand)} \\ 3x + y &\leq 3 \quad \text{(Capacity)} \\ x, y &\geq 0 \quad \text{(Non-negative)} \end{aligned}$$

– **Graphical Representation**



– **Solution Steps**

1. Find boundary lines:
   * Demand: $2x + 3y = 6$
   * Capacity: $3x + y = 3$
2. Find intersection point: From $y = 3 - 3x$ and $2x + 3y = 6$:

$$\begin{aligned} 2x + 3(3 - 3x) &= 6 \\ 2x + 9 - 9x &= 6 \\ -7x &= -3 \\ x &= \frac{3}{7} \approx 0.43 \end{aligned}$$

   This gives $y = \frac{12}{7} \approx 1.71$
3. Corner points: $(0, 2)$ and $(\frac{3}{7}, \frac{12}{7})$

4. Evaluate objective at corners:
   * At $(0, 2)$: $Z = 4(0) + 5(2) = 10$
   * At $(\frac{3}{7}, \frac{12}{7})$: $Z = 4(\frac{3}{7}) + 5(\frac{12}{7}) = \frac{72}{7} \approx 10.29$

– **Result**

The minimum cost is $\boxed{Z = 10}$ at $\boxed{(0, 2)}$.

So we should produce 0 units of product X and 2 units of product Y.

Two-Phase Method

The standard form:

$$\text{Max} \quad Z' = -4x - 5y + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot a_1$$

Subject to

$$2x + 3y - s_1 + 0 \cdot s_2 + a_1 = 6$$
$$3x + y + 0 \cdot s_1 + s_2 + 0 \cdot a_1 = 3$$
$$x, y, s_1, s_2, a_1 \geq 0$$

Auxiliary objective function:

$$\text{Max} \quad z^* = 0 \cdot x + 0 \cdot y + 0 \cdot s_1 + 0 \cdot s_2 - a_1$$

Initial basic feasible solution:

$$a_1 = 6, \quad s_2 = 3, \quad x = y = s_1 = 0$$

PHASE - I

CB_i	c_j / BV	0 / x	0 / y	0 / s_1	0 / s_2	-1 / a_1	b_j	b_j/a_{ij}
		a_{ij}						
-1	a_1	2	$\boxed{3}$	-1	0	1	6	2
0	s_2	3	1	0	1	0	3	3
$z_j = \Sigma(CB_i \times a_{ij})$		-2	-3	1	0	-1		
$c_j - z_j$		2	3	-1	0	0		
0	y	2/3	1	-1/3	0	×	2	
0	s_2	7/3	0	1/3	1	×	1	
z_j		0	0	0	0	×		
$c_j - z_j$		0	0	0	0	×		

$c_j - z_j \leq 0$, and no artificial variable appears in the basic solution. Therefore, an optimal BFS to the auxillary LPP has been attained. Hence, we move onto PHASE II.

PHASE-II

		a_{ij}					
CB_i	c_j	-4	-5	0	0	b_j	b_j/a_{ij}
	BV	x	y	s_1	s_2		
-5	y	2/3	1	-1/3	0	2	
0	s_2	7/3	0	1/3	1	1	
$z_j = \Sigma(CB_i \times a_{ij})$		-10/3	-5	5/3	0		
$c_j - z_j$		-2/3	0	-5/3	0		

Since all $c_j - z_j \leq 0$, we have an optimal solution

$x = 0$, $y = 2$.

Min $Z = 4(0) + 5(2) = 10$ //

Comparison, Advantages & Conclusion

5.1 Comparison of Graphical and Algebraic Methods

Linear Programming Problems (LPPs) can be solved using both graphical and algebraic methods. The graphical method provides a visual and intuitive approach suitable for problems involving two variables. In contrast, the algebraic method, such as the Two-Phase Simplex Method, extends to higher-dimensional problems and provides more precise, scalable solutions.

In the context of this project, both methods were used to validate the model and verify accuracy. The graphical method offered clarity in understanding the feasible region and optimal point, whereas the Two-Phase method delivered a robust, systematic approach suitable for real-world applications.

Table 1: Comparative Analysis of Graphical and Algebraic Methods

Feature	**Graphical Method**	**Algebraic (Two-Phase) Method**
Scalability	Limited to 2 variables	Suitable for any number of variables
Visualization	Provides a clear visual understanding	Lacks visual interpretation
Complexity	Simple and intuitive	Involves matrix operations
Accuracy	Approximate (depends on plotting)	Precise and algebraic
Real-world applicability	Mostly educational	Widely used in industry

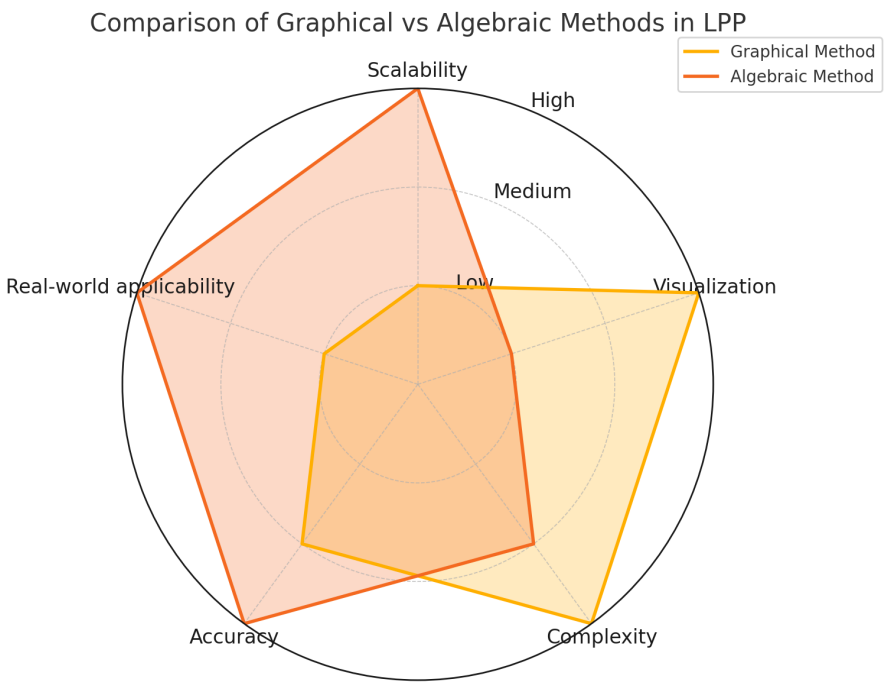


Figure 2: Radar chart comparing the Graphical Method and Algebraic (Two-Phase) Method in LPP across five key attributes — Scalability, Visualization, Complexity, Accuracy, and Real-world Applicability.

5.2 Advantages and Disadvantages of Each Method

The graphical method is advantageous for learning and initial problem understanding. It offers visual insights into feasible regions and constraints boundaries. However, it becomes impractical for problems involving more than two variables or complex constraints.

On the other hand, the Two-Phase method is computationally more intensive but remains reliable and adaptable to complex optimization scenarios. It enables rigorous handling of artificial variables and infeasible starting points.

Advantages of Graphical Method:

- Easy to use for small-scale problems.
- Good for visual learners and conceptual understanding.

Disadvantages of Graphical Method:

- Not applicable to higher-dimensional problems.
- Limited accuracy in manual plotting.

Advantages of Two-Phase Method:

- Applicable to any number of variables and constraints.
- Provides accurate and algebraically verifiable solutions.

Disadvantages of Two-Phase Method:

- Requires knowledge of matrix algebra.
- Involves iterative calculations or programming tools.

5.3 Justification of Method Chosen

Given the simplicity of the formulated transportation cost problem (with only two decision variables), the graphical method was suitable for preliminary analysis. However, for formal validation and scalability, the Two-Phase Simplex method was chosen as the definitive approach. It not only confirmed the results obtained graphically but also demonstrated the extensibility of the model to larger, more complex logistics scenarios.

5.4 Real-life Applications of LPP in Traveler Coordination

Linear programming plays a pivotal role in travel coordination and cost optimization. Applications include:

- Route optimization for fleet management in travel agencies.
- Ride allocation for ride-sharing platforms to minimize wait time and fuel costs.
- Tour scheduling to manage cost and time across multiple destinations.
- Resource allocation in public transport systems to optimize capacity and minimize operational expenditure.

According to studies such as those published in *Elsevier's Transportation Research* and *Springer's Logistics Series*, LPPs have improved logistical efficiency by 15–30% in multimodal transport networks.

5.5 Challenges Faced and How They Were Overcome

Throughout the project, several challenges were encountered:

- Constraint formulation was initially vague, requiring iterative clarification of the real-world situation.
- Graphical precision in plotting feasible regions was limited, resolved by using digital graphing tools.
- Understanding artificial variables in the Two-Phase method was challenging and overcome through theoretical study and problem sets.
- Software limitations were mitigated by cross-validating solutions using Python-based solvers and Excel tools.

These challenges strengthened the team's understanding of real-world optimization modeling and problem-solving frameworks.

5.6 Key Learnings from the Project

This research project facilitated a deep understanding of:

- Mathematical modeling of practical problems using LPP.
- Comparative strengths of different solution methods.
- The importance of accurate constraint formulation and objective function design.
- The practical utility of optimization tools in transportation and logistics.

The collaborative effort also improved analytical thinking, technical communication, and the ability to transition from theoretical concepts to real-world applications.

5.7 Future Scope and Enhancements

Future enhancements to this research can include:

- Integration with APIs (e.g., Google Maps, Uber API) to gather live data on routes, fuel cost, and time.
- Extension to multi-objective optimization, combining cost, time, and environmental sustainability.
- Development of a software tool that automatically formulates and solves LPPs for travel coordination.
- Incorporation of uncertainty modeling using stochastic linear programming to handle delays, breakdowns, or demand surges.

This foundational work opens avenues for more advanced applications in smart transport, sustainable tourism, and AI-based route planning.